VI-Based Deep Uncertainty Estimation

- We saw, in general, how to model Epistemic Uncertainty in a Bayesian way
- We know that it is generally interactable
- We learned about one important approximation technique called variational inference



Example 0

Sanity Check!

1. Have some training data $\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$
2. Assume a prior distribution $P(\boldsymbol{\theta})$
   $P(\boldsymbol{\theta}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$
3. Design a likelihood function $P(y|\mathbf{x}, \boldsymbol{\theta})$
   $P(y|\mathbf{x}, \boldsymbol{\theta}) = f_{NN}(\mathbf{x}; \boldsymbol{\theta})|_y$ --a softmax neural network
4. Decide on approximating distribution $Q_{\boldsymbol{\omega}}(\boldsymbol{\theta})$
   $Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) = \delta_{\boldsymbol{\omega},\boldsymbol{\theta}}$
5. Find $\boldsymbol{\omega}$ that optimizes the following objective

$$\min_{\boldsymbol{\omega}} \sum_{i=1}^{n} H(\delta_{y,y_i}, f_{NN}(\mathbf{x}_i; \boldsymbol{\omega})) + \frac{\|\boldsymbol{\omega}\|^2}{2}$$

$$\max_{\boldsymbol{\omega}} \sum_{i=1}^{n} \log f_{NN}(\mathbf{x}_i; \boldsymbol{\omega})\Big|_{y_i} - \frac{\|\boldsymbol{\omega}\|^2}{2}$$

$P(\boldsymbol{\theta}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$

$Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) = \delta_{\boldsymbol{\omega},\boldsymbol{\theta}}$

$Q_{\boldsymbol{\omega}}$ is a discrete distribution

$$\max_{\boldsymbol{\omega}} \sum \left(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) \sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}) \right) - D_{KL}\big(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) \,\|\, P(\boldsymbol{\theta})\big)$$